



NATIONAL OPEN UNIVERSITY OF NIGERIA
PLOT 91, CADASTRAL ZONE, NNAMDI AZIKIWE EXPRESSWAY, JABI-ABUJA
FACULTY OF COMPUTING
DEPARTMENT OF COMPUTER SCIENCE
2025_2 EXAMINATIONS

COURSE CODE: CIT314

COURSE TITLE: Computer Architecture and Organization

CREDIT: 3 Units

TIME: 3 Hours

INSTRUCTION: Answer Question ONE (1) and any other THREE (3) Questions

Question 1

1a. Distinguish between Computer Organization and Computer Architecture. **(5 Marks)**

1b. List three (3) advantages of Hardwired Control Units. **(3 Marks)**

1c. List five (5) characteristics of Magnetic Storage Tapes. **(5 Marks)**

1d. In computer architecture, data needs to be accessed from the memory for various purposes. Identify and explain four (4) methods used to access memory. **(12 Marks)**

Question 2

2a. List four (4) primary types of Auxiliary Storage Devices. **(2 Marks)**

2b. Highlight four (4) Advantages of Micro-programmed Control Unit. **(4 Marks)**

2c. Identify and describe four (4) access rights of interest that are used for protecting the programs residing in memory. **(9 Marks)**

Question 3

3a. Multiprogramming operating system allows executing multiple processes by monitoring their process states and switching in between processes. Outline three (3) Advantages and three (3) Disadvantages of Multiprogramming. **(6 Marks)**

3b. Identify and describe two (2) main types of perimeter-based technologies. **(3 Marks)**

3c. Compare Synchronous versus Asynchronous Data Transmission in four (4) sentences. **(6 Marks)**

Question 4

4a. Discuss the following two (2) basic characteristics of Fault Tolerant Systems. **(4 Marks)**

- i. No Single Point of Failure
- ii. Fault isolation or identification

4bi. The concepts of faults, errors, and failures can be best presented by the use of a Three-Universe Model that is an adaptation of Four-Universe Models; Briefly describe the First Universe. **(2 Mark)**

4bii. Fault Tolerance methods can be applied to cloud computing in three (3) levels. Identify and explain these levels. **(4 $\frac{1}{2}$ Marks)**

4c. Discuss three (3) Fault tolerance design goals. **(4 $\frac{1}{2}$ Marks)**

Question 5

5a. Strobe control method of data transfer uses a single control signal for each transfer. Using a timing diagram, describe a Source Initiated Strobe. **(3 Marks)**

5b. Several types of faults may occur in the cloud environment leading to failures and performance degradation. Identify and describe four (4) main types of faults in cloud computing environments. **(6 Marks)**

5c. Identify and explain four (4) fault tolerance techniques or methods often used by programmers. **(6 Marks)**

Question 6

6a. Highlight three (3) drawbacks that occur in virtual memories. **(3 Marks)**

6b. In the context of memory organization, discuss the concept of Access time (latency). **(2 Marks)**

6c. Discuss the key differences between Hardwired Control Unit and Microprogrammed Control Unit by filling in the blank spaces in the table below. **(10 Marks)**

ATTRIBUTES	HARDWIRED CONTROL UNIT	MICROPROGRAMMED CONTROL UNIT
1. Speed		
2. Cost of Implementation		
3. Flexibility		
4. Ability to Handle Complex Instructions		
5. Decoding		
6. Applications		
7. Instruction set of Size		
8. Control Memory		
9. Chip Area Required		
10. Occurrence		



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2025_1 EXAMINATION...

COURSE CODE: CIT312

COURSE TITLE: Computer Architecture and Organization

CREDIT: 3 Units

TIME: 3 Hours

INSTRUCTION: Answer Question ONE (1) and any other THREE (3) Questions

Question 1

- 1a. Highlight two (2) Disadvantages of Microprogrammed Control Unit. **(1 Mark)**
- 1b. Discuss by highlighting seven (7) key points about Magnetic Disk Storage. **(7 marks)**
- 1c. Examine Five (5) known limitations of Asynchronous Controllers. **(5 Marks)**
- 1d. Discuss how fault-tolerant architectures can be used to address circuit reliability concerns. **(12 marks)**

Question 2

- 2a. Examine the core components of a critical system to improve fault tolerance. **(4 marks)**
- 2b. Discuss cache memory in computer architecture. **(5 Marks)**
- 2ci. Define Registers and describe two forms of registers. **(4 Marks)**
- 2cii Give two (2) examples of Registers you defined in (i) and highlight their function. **(2 Marks)**

Question 3

- 3a. Examine when Replacement Algorithm is useful. **(2 marks)**
- 3b. The **one-address** instruction **ADD R1** = the **three-address** instruction **ADD R1,Racc,Racc** or to the **two-address** instruction **ADD R1,Racc**. How? **(3 marks)**
- 3c. Discuss the basic characteristics of fault tolerance systems. **(10 marks)**

Question 4

- 4a. Assemblers are programs that generate machine code instructions from a source code program written in assembly language. Highlight the functions of assemblers. **(3 Marks)**
- 4b. Discuss the issues associated with hardware and software fault tolerance. **(5 marks)**
- 4c. With the aid of a well-labelled diagram, discuss the process of **Handshaking** in Asynchronous buses. **(7 Marks)**

Question 5

- 5a. What is a Redundancy in fault tolerant systems? **(2 marks)**
- 5b. Highlight the components of the 16-bit registers in a simple accumulator-based processor. **(4 Marks)**
- 5c. Discuss three (3) main methods through which data can be transferred to and from the peripherals. **(9 Marks)**

Question 6

- 6a. Identify two basic techniques to increase the instruction execution rate of a processor. **(2 Marks)**
- 6b. Highlight three (3) motivations of parallel processing. **(3 marks)**
- 6c. Discuss two (2) forms of redundancy in fault tolerant systems. **(10 marks)**



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2024_2 EXAMINATION

CIT 314: COMPUTER ARCHITECTURE AND ORGANIZATION **Credit: 2½ units**

TIME: 2 1/2 HOURS

INSTRUCTION: Answer Four (4). Question one is Compulsory

1a. The development of fault models is an essential part of the process in determining the reliability of a system. Examine four major goals exist when devising a fault model. (6 marks)

1b. Discuss the concept of protecting paging, highlight the merits and demerits of paging. (8 marks)

1c. With the aid of well-labelled diagram, describe two-phase communication protocol in asynchronous signaling conventions (6 marks)

1d. Briefly discuss the following in asynchronous circuits

h) Timing model **(3 marks)**

ii) Mode **(2 marks)**

2a. What is the key benefit of fault tolerance?(3 marks)

2b. Examine other mapping process similar to paging system. (4 marks)

2c. Highlight the operations of floppy disk . **(8 marks)**

3a. One of the properties of logical space is that it uses variable-length segments. The length of each segment is allowed to grow and contract according to the needs of the program being executed. Describe one way of specifying the length of a segment. (2 marks)

3b. Examine the recording method of floppy disks **(6 marks)**

3c. Describe the benefits of Asynchronous Control (7 marks)

4a. State and discuss internal registers being used by the CPU to exchange data with memory. **(4 marks)**

4b. Describe the function of the following in input/output operation: **(5 marks)**

- i. Program Counter (PC)
- ii. Instruction Register (IR):
- iii. Memory Address Register (MAR):
- iv. Memory Buffer Register:
- v. I/O address register (I/OAR):

I/O buffer register (I/OBR):

4c. With the aid of well-labelled diagram, describe the components of hard drives. **(7 marks)**

5a. Distinguish the external and internal drives.(4 marks)

5b. Describe a programming with parallel processing (5 marks)

5c. The directory structure records information such as name, location, size and type for all files stored on the storage devices. Highlight the issue(s) associated to single-Level directory and discuss solution provided by other directory structures **(6 marks)**

6a. List four examples of Asynchronous Communication. (2 marks)

6b. Examine the merits and demerits of multiprogramming in computer architecture. (6 marks)

6c. CD-ROM disks are one of the storage disks made of reflective metals and written during the process of manufacturing by high power laser beam. Highlight the characteristics of CD=ROM. (7 marks)



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2024 1 EXAMINATION

CIT 314: COMPUTER ARCHITECTURE AND ORGANIZATION **Credit: 2½ units**

INSTRUCTION: **Answer Four (4). Question one is Compulsory**

- 1a. Highlight two disadvantages of Microprogrammed Control Unit (5 mark)
- 1b. Discuss by highlighting, the storage location of a magnetic disk (10 marks)
- 1c. Examine limitations of Asynchronous Controllers. (10 marks)

- 2a. Examine core components of a critical system to improve fault tolerance. (4 marks)
- 2b. Discuss cache memory in computer architecture. (5 marks)
- 2c. Define registers and discuss two forms of registers with examples. (6 marks)

- 3a. Examine when replacement algorithm is useful. (2 marks)
- 3b. The **one-address** instruction **ADD R1** = the **three-address** instruction **ADD R1,Racc,Racc** or to the two-address instruction **ADD R1,Racc**. How? (3 marks)
- 3c. Discuss the characteristics of fault tolerance systems. (10 marks)

- 4a. Assemblers are the programs that generate machine code instructions from a source code program written in assembly language. Highlight the functions of assemblers (3 Marks)
- 4b. What are the issues associated to Hardware and software tolerant. (5 marks)
- 4c. With the aid of well-labelled diagram, discuss the process of **Handshaking in Asynchronous buses** (7 Marks)

- 5a. What is a redundancy in fault tolerant? (2 marks)
- 5b. Highlight the components of the 16-bit registers in a simple accumulator-based processor (4 Marks)
- 5c. Discuss methods through which data can be transferred to and from the peripherals. (9 marks)

- 6a. List two basic techniques to increase the instruction execution rate of a processor
- 6b.** The quest for higher-performance digital computers has brought about the development of parallel processing. Highlight the motivations of parallel processing. (3 marks)
- 6c. Discuss forms of redundancy in fault tolerant. (10 marks)



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2023_2 EXAMINATIONS

CIT 314: COMPUTER ARCHITECTURE AND ORGANIZATION **Credit: 2½ units**

INSTRUCTION: Answer Four (4). Question one is Compulsory

- 1a. Examine the merits and demerits of CD-ROMs. (10 marks)
- 1b. In a segmented-page organization, each entry in the page table may have its own protection field to describe the access rights of each page. Identify some of the access rights of interest that are used for protecting the programs residing in memory. (4 marks)
- 1c. Juxtapose the Synchronous and Asynchronous Transmission. (5 marks)
- 1d. The cause-effect relationship implied in the three-universe model leads to the definition of two important parameters; fault latency and error latency. Highlight the characteristics of faults. (6 marks)
- 2a. Discuss memory mapping and its important in computer architecture (4 marks)
- 2b. Examine drawbacks in the virtual memory and important properties for maintaining consistency in the memory hierarchy. (7 marks)
- 2c. Describe sources of faults you understand in fault tolerant. (4 marks)
- 3a. Virtual memories enable processes in the systems to share CPU and main memory with other processes. Highlight the importance of virtual memories. (3 marks)
- 3b. In computer architecture, it is the responsibility of the control Unit to tell the computer's memory, arithmetic/logic unit and input and output devices how to respond to the instructions that have been sent to the processor. Examine the functions of a control unit. (6 marks)
- 3c. Describe the following in fault tolerant. (6 marks)
 - I) Nature of a faults
 - II) Fault Duration
 - III) Permanent fault
 - IV) Transient fault

- v) Intermittent fault
- vi) Fault Extent

4a. Define fault tolerance analysis. (1 mark)

4b. What is the goal of fault tolerant? (2 marks)

4c. Write short note on the following Compact Disk-Read Only Memory (12 marks)

- i) Read-only memory (ROM) disks (2 marks)
- ii) Write-once read-many (WORM) disks (2 marks)
- iii) Re-writeable, write-many read-many (WMRM) disks (2 marks)
- iv) WORM (write once, read many) (2 marks)
- v) Erasable Optical Disk: (2 marks)
- vi) Touchscreen Optical Device: (2 marks)

5a. Distinguish between machine and assembly languages (1 Mark)

5b. Highlight the basic steps for READ and WRITE memory operations (4 marks)

5c. Discuss the goals of fault tolerant (10 marks)

6a. Discuss Fault tolerance systems. (5 marks)

6b. Examine types of control unit in central processing unit. (4 marks)

6c. With the aid of examples, describe three address instruction in computer architecture (6 marks)



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2022_1 EXAMINATIONS

CIT314 – Architecture and Organization II

Credit: 3 units

TIME ALLOWED: 2½ Hours

INSTRUCTION: Answer Question 1 and any other THREE (3) Questions

QUESTION ONE

a) Write short notes on each of the following

- i. ROM
- ii. RAM
- iii. Computer Architecture **(6 marks)**

b) Explain the concept of the von Neumann computer. **(5 marks)**

c) State the sequence of operations of the control unit in one clock pulse. **6 marks**

d) Distinguish the difference between computer organization and computer function. **(5 marks)**

e) State the first three (3) single instruction cycle **(3 marks)**

QUESTION TWO

a. Explain any four important instruction set design issues in computer architecture. **(6 marks)**

b. Briefly explain the key components of a Mainframe Symmetric system **(6 marks)**

c. Explain Instruction Address Calculation (ac) **(3 marks)**

QUESTION THREE

a. Evaluate the following using two's complement, expressing your answer in binary

i. $17_{10} - 10_{10}$

ii. $9_{10} + 11_{10}$

(8 marks)

b. Outline the four (4) categories of micro-operations known to you **(4 marks)**

c. State three (3) basic functions that a computer can perform. **(3 marks)**

QUESTION FOUR

a. Explain the difference between hardwired control and micro programmed control.
(6 Marks)

- b. Enumerate the characteristics of RISC Architectures. **(6 Marks)**
- c. All instructions follow same sequence of five pipeline stages. Mention two (2). **(2 marks)**

QUESTION FIVE

- a. Differentiate between the following terms:**(6 marks)**
 - i.) Control bus
 - ii.) Address bus
 - iii.) Data bus
 - iv.) Karnaugh map
 - v.) Boolean algebra
 - vi.) Tag check
- b. List and explain any three performance parameters used in memory **(6 marks)**
- c. State and explain the two (2) objectives of Operating System(OS) **(3 marks)**

QUESTION SIX

- a. Evaluate the following using two's compliment **(5 marks)**
 - i. 0000 0101 – 0000 1000
 - ii. 0000 0111 – 0000 0011
- b. With the aid of an illustrative diagram, explain the five-state process model. **(10 marks)**



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2023_1 POP EXAMINATION₂₁

CIT314 – Architecture and Organization II

Credit: 3 units

TIME ALLOWED: 2½ Hours

INSTRUCTION: Answer Question 1 and any other THREE (3) Questions.

Question One

- a. Discuss the types of faults may occur in the cloud environment leading to failures and performance degradation. (4 marks)
- b. Outline the challenges of implementing fault tolerance in cloud computing. (7 marks)
- c. With example describe the concept of cloud computing. (5 marks)
- d. List the major issues that may arise in the process of fault modelling and evaluation. (5 marks)
- e. Describe the concept of virtual memory as a tool for caching. (4 marks)

Question Two

- a. Describe any four important instruction set design issues in computer architecture. (6 marks)
- b. Clearly highlight, the advantages and disadvantages of pipelining. (5 marks)
- c. Define the following terms:
 - a. Translation lookaside buffer (TLB) (2 marks)
 - b. The Second Chance Page Replacement Algorithm. (2 marks)

Question Three

- a. Explain the architectural concept of a computer system that enables it operate such that all programs are executing at the same time. (5 marks)
- b. Give a short description of floppy disks recording methods. (6 marks)
- c. Why is virtual memory important to a programmer? (2 marks)
- d. Define Fault Tree Analysis. (2 marks)

Question Four

- a. Highlight the major goals of devising a fault model. (4 marks)
- b. Discuss the term Fault Models (6 marks)
- c. What is virtual memory and outline its' important capabilities that actuates memory abstraction? (5 marks)

Question Five

- a. With the aid of a diagram, discuss the hierarchy design of memory in a computer system. (6 marks)
- b. List the main characteristics of magnetic tape memory. (5 marks)
- c. Why is memory hierarchy relevant in the organization of a computer system? (4 marks)

Question Six

- a. Describe the possible ways of Data transfer to and from the peripherals. (6 marks)
- b. Discuss the features inherent in Compact Disk Read-Only Memory. (7 marks)
- c. What are the drawbacks of using virtual memory? (2 marks)



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2022_2 EXAMINATIONS

CIT314 – Architecture and Organization II

Credit: 3 units

TIME ALLOWED: 2½ Hours

INSTRUCTION: Answer Question 1 and any other THREE (3) Questions

QUESTION ONE

- a. Why is virtual memory important to a programmer? (2 marks)
- b. Describe any four important instruction set design issues in computer architecture. (6 marks)
- c. Explain the difference between hardwired control and micro programmed control. (6 Marks)
- d. Give a short description of floppy disks recording methods. (6 marks)
- e. State the function of the following terms: (5 marks)
 - i. Control bus
 - ii.) Address bus
 - iii.) Data bus
 - iv.) Karnaugh map
 - v.) Boolean Algebra

QUESTION TWO

- a. Discuss the major advantages of a magnetic tape. (4 marks)
- b. Enumerate the characteristics of RISC Architectures. (6 Marks)
- c. With the aid of diagram, describe the basic feature of Sequential Access Method. (3 marks)
- d. Define the term Computer Architecture 2 Marks

QUESTION THREE

- a. Discuss the three performance parameters used in memory (6 marks)
- b. Evaluate the following using two's complement: (5 marks)
 - a. 0000 0101 – 0000 1000
 - b. 0000 0111 – 0000 0011

- c. Briefly describe the essential features of an auxiliary memory. (4 marks)

QUESTION FOUR

- a. Briefly explain the following terminologies:
- i.) Processor instruction set (*1 mark*)
 - ii.) Opcodes (*1 mark*)
 - iii.) Arithmetic and Logic Unit (ALU) . (*1 mark*)
 - iv.) Two's complement notation *2 marks*
 - v.) Memory buffer register (MBR) *1 mark*
- b. Write short-note on perimeter-based technologies. (4 marks)
- c. Identify the important capabilities of virtual memory. (3 marks)
- d. Mention four (4) instructions that follow same sequence of five pipeline stages. (*2 marks*)

QUESTION FIVE

- a. Write short notes on each of the following
- i. ROM
 - ii. RAM
 - iii. Computer Architecture (*6 marks*)
- b. The operation of the processor is determined by the instructions it executes. Enumerate four Main elements of a machine instruction. (*6 marks*)
- c. State the sequence of operations of the control unit in one clock pulse. *3 marks*

QUESTION SIX

- a. What is memory-mapping, and identify the principal benefits to computing? (*6 marks*)
- b. The main memory is the central storage unit in a computer system. Write a short note to discuss the infrastructure of main memory in a computer system. (*7 marks*)
- c. Discuss the two (2) main objectives of Operating System (*2 marks*)

